Do Now · retrieval

Lesson 080 · Insertion Sort: Grow the Sorted Prefix

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What gets fixed after a bubble pass?

Q2

What does insertion-sort key mean?

Q3

What does sorted prefix mean?

Q4

BST root10 with right child15: insert key12. Show comparisons and position.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Understand — The left portion is kept sorted. Read the next item as key and compare backwards with earlier elements.

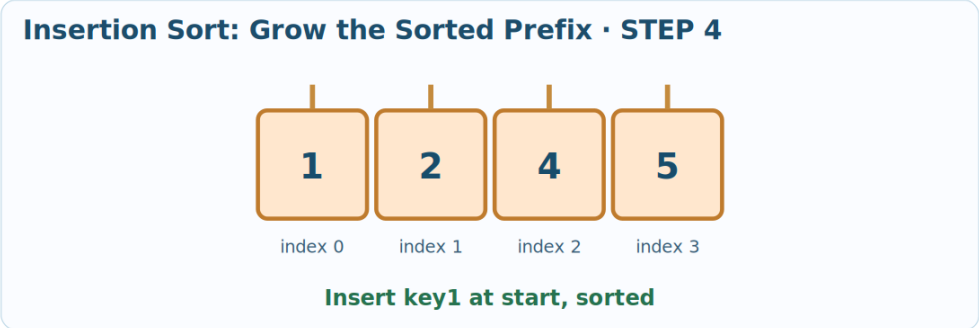
Core idea 2

Trace — Shift larger elements one position to the right, then insert key in the vacant position.

Core idea 3

Explain — Worst $O(n^2)$, already-sorted best $O(n)$, auxiliary $O(1)$; especially useful on small/nearly sorted arrays.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

insertion sort — Insert key into ordered prefix.

key — Value currently being inserted.

shift — Move existing value to make room.

sorted prefix — Already-sorted left section.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

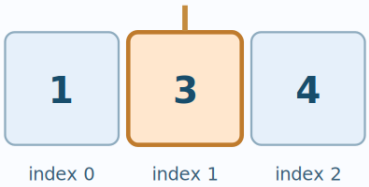
Insertion-sort [4,1,3]; give states after each key.

Follow the worked method

1. Check the working — Key1→[1,4,3]; key3→[1,3,4].

Source model · final illustrated step

IDO · TRACE



Key1→[1,4,3]; key3→[1,3,4].

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Check the working
Shift5 and3 right then write2 at index0, giving [2,3,5].

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Give arrays after first two key insertions into [5,2,4,1]. [6]

Question 2

Complete insertion of final key1 into [2,4,5,1], showing working. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Compare insertion and bubble sort method/time on already sorted input. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Insertion sort always requires a separate second array to store sorted values.

Use key vocabulary from memory

1. Explain insertion sort in your own words.

2. Explain key in your own words.

3. Explain shift in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.